TITLE: Sleuth the crook (A story based game)

AIM: To create an interactive, murder mystery based story game using the programming language C++ with the help of software TurboC++

OBJECTIVES:

- To create a title page which is visible first, on opening the game.
- To create a screen where user input is required to start the game (using if-else conditional statements)
- To create an original storyline based on a forensics murder mystery.
- To use if-else conditional statements for displaying different outputs based on user inputs.
- To use switch case nested within if-else conditional statements for displaying different outputs based on user inputs.
- To calculate score based on user input.
- To create a final result and create a score board to assess performance.
- To create a final 'Thank you for playing' screen.

EXPECTED OUTPUT:

- To obtain a title screen with an aligned title.
- To display an 'Are you ready to play?' screen with options 'Yes' and 'No'.
- To obtain the murder mystery storyline and questions based on the story.
- To take in user input and display further questions as per the input of the user.
- To calculate the score after every question as per user input.
- To display the result as per user's input
- To display the calculated score which depends on the choices that the user made.
- To display the score board below the calculated score.
- To display a 'Thank you for playing' screen, at the end of the game.
- To change the colour of the background screen for the end result.
- To insert graphics in the game.
- To add a timer to the questions.

OBSERVED RESULTS:

- A title screen with an aligned title was obtained.
- ‘Are you ready to play?’ screen with options ‘Yes’ and ‘No’ was displayed.
- A murder mystery storyline and questions based on the story were created.
- User input was taken and further questions were displayed as per the input of the user.
- The score was calculated after every question as per user input.
- The result was displayed as per user’s input
- The calculated score was displayed which depended on the choices that the user made.
- The score board was displayed below the calculated score.
- A ‘Thank you for playing’ screen was displayed, at the end of the game.

WORKFLOW:

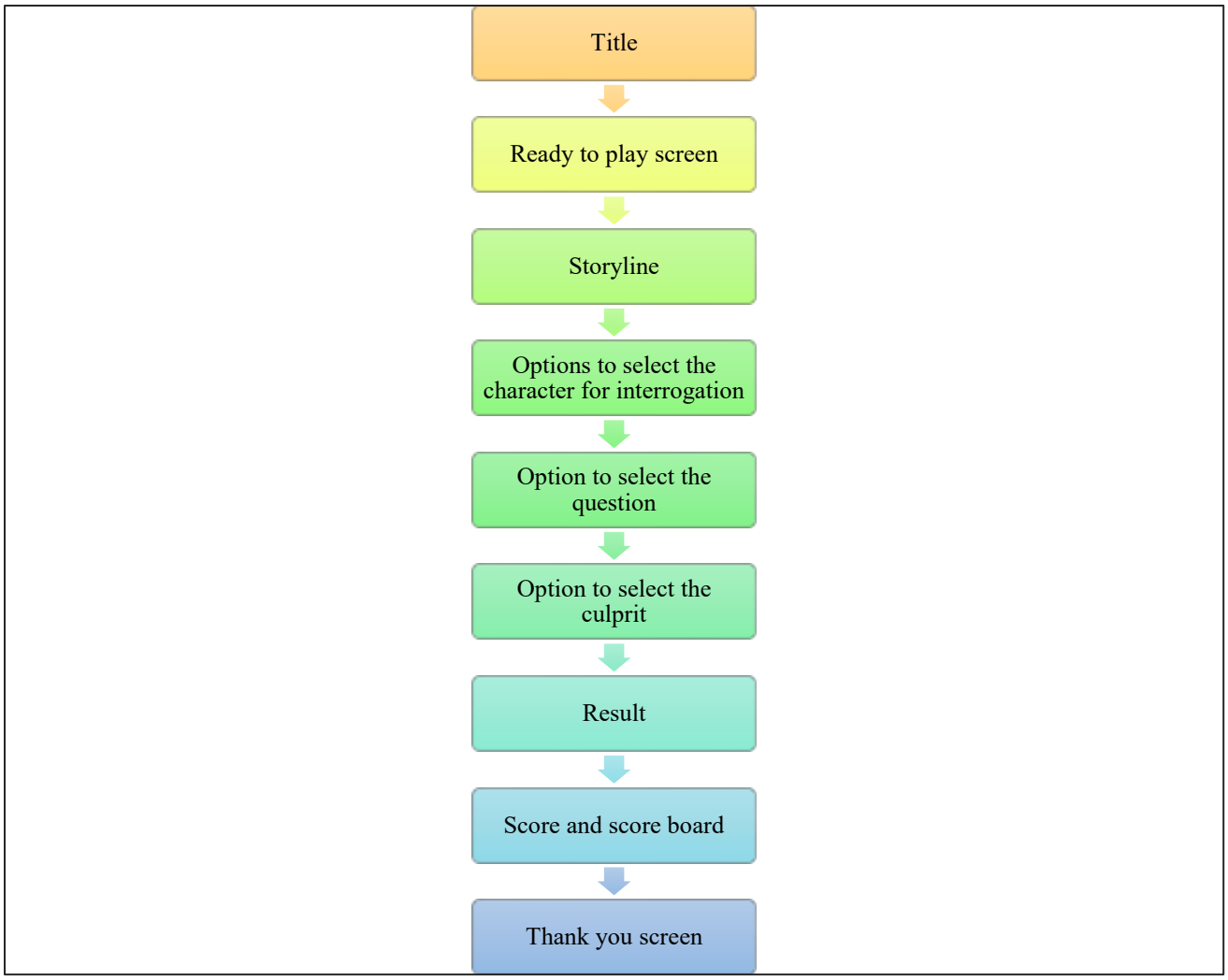

Fig 1: Flowsheet displaying the entire workflow of the game

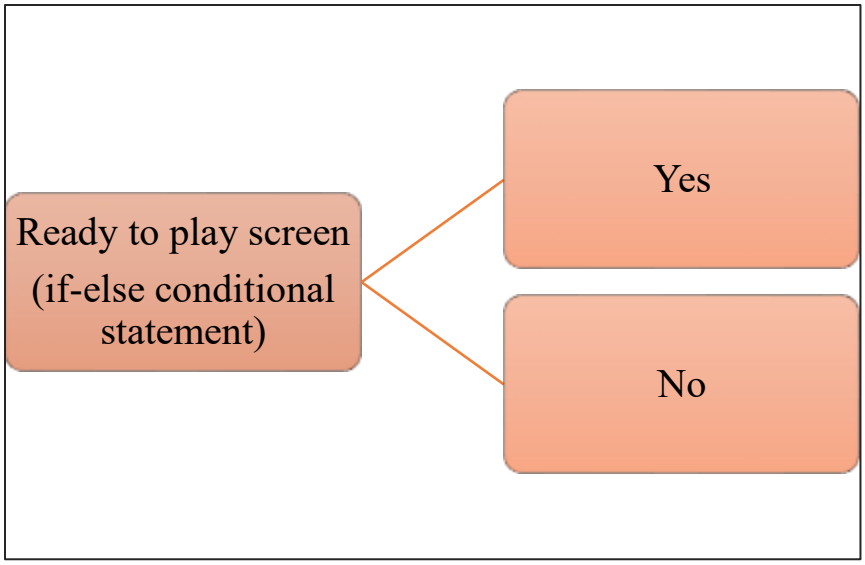

Fig 2: Diagram to show workflow for ‘Are you ready to play?’ screen

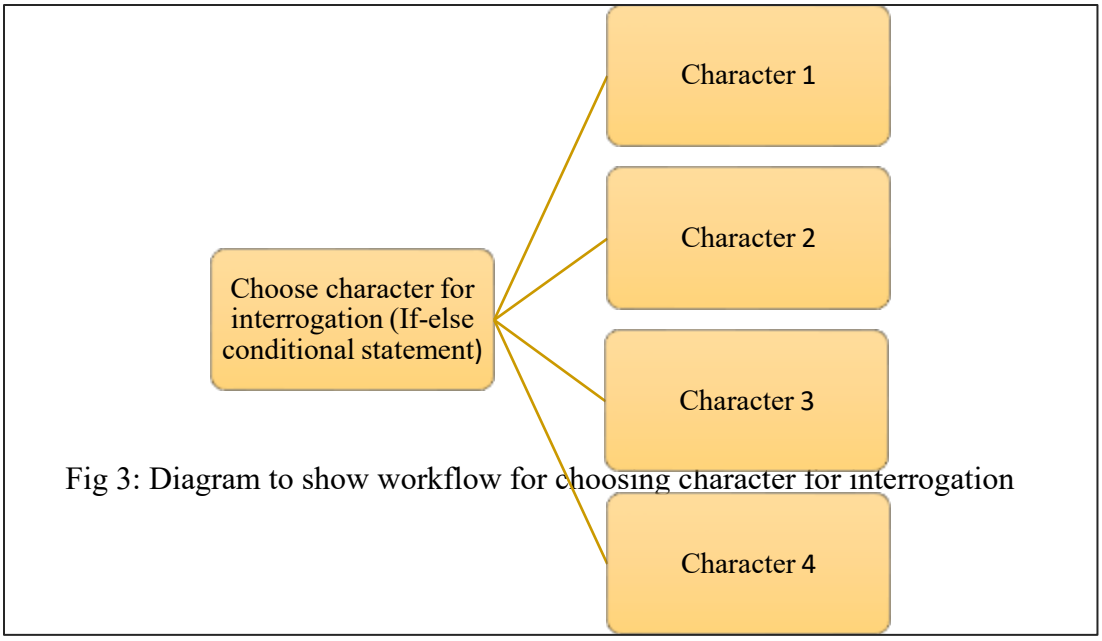

Fig 3: Diagram to show workflow for choosing character for interrogation

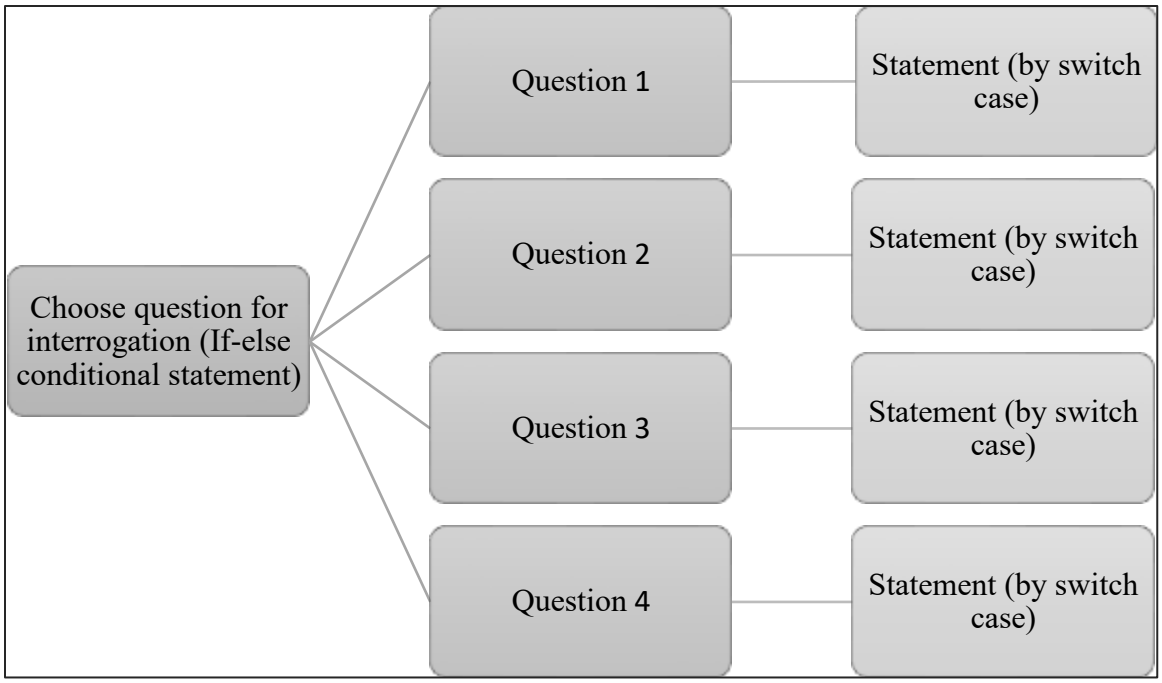

Fig 4: Diagram for showing workflow to choose question for interrogation.

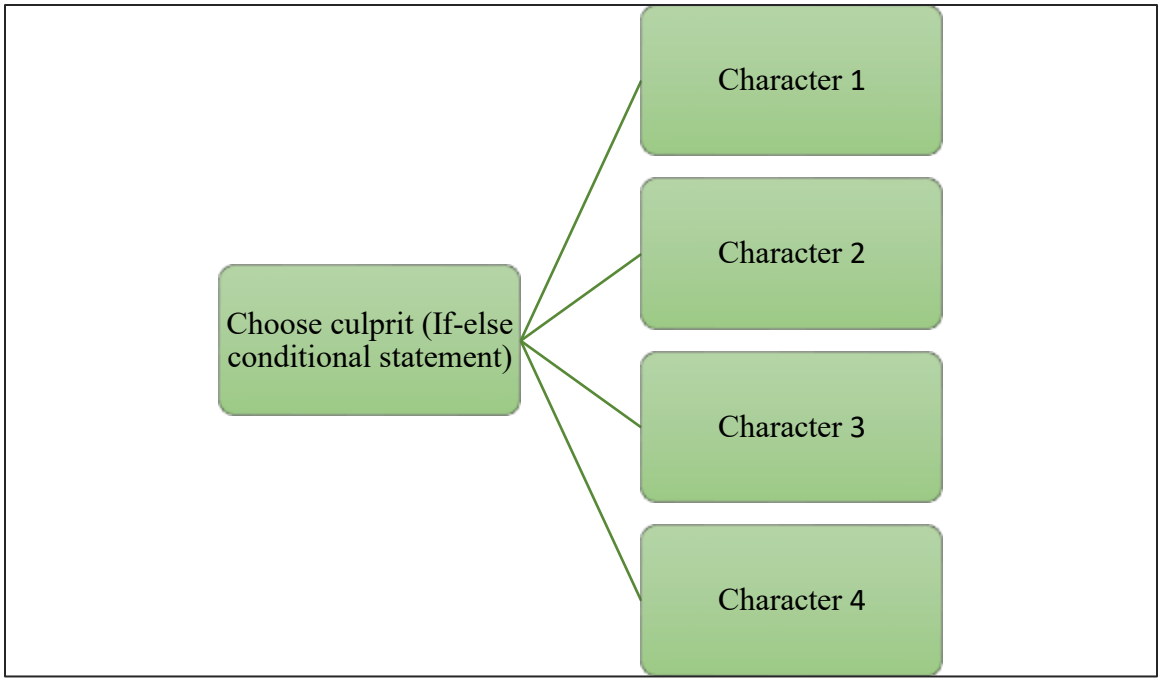

Fig 5: Diagram to show workflow to choose culprit

CODE:

```
#include<iostream.h>

#include<conio.h>

int main()

{clrscr();

int a, b, score=0, l;

char Y, N, ans;

cout<< "\n\n\n\n\n\n\n\n\n\n\t\t\t\t SLEUTH THE CROOK \n";

getch();

clrscr();

cout<< "\n\n\n\n\n\n\n\n\t\t\t ARE YOU READY TO PLAY? \n ";

cout<< "\n\t\t Press Y to continue \t";

cout<< " Press N to leave \t";

cin>>ans;

clrscr();

if(ans=='N')

 {

 cout<< " Thank you for playing\n ";

 }

else

 {

 cout<< "\n\n\n\n\n\n\n\t\t\t\t Choose topic: \n ";

 cout<<"\n\t\t\t Case 1. The author's murder\n" ;

 cin>>a;

 clrscr();
```

```
}
        cout<<" You chose case 1 \n";
        cout<< " Hello detective. Here is your case: \n\n";
cout<<" At 3:44 pm on a Sunday, first responders received a call that Rick, an author, was found
        dead in his living room.";
cout<< "There was no physical trauma to his body on a preliminary assessment.";
cout<< " The living room was found in shambles, with an upended sofa and stains on the rug.";
cout<< " 2 glasses with water in them were found on a nearby table.";
cout<< " His latest book was found torn near the table.";
cout<< " A broken vase laid near the window sill.";
cout<< " There was nobody else in the house, apart from his dog, which had been found eating
        peanut butter from the jar.";
cout<< " The call had been made by his househelp, Stacy, when she let herself in the house for
        cleaning.";
cout<< " An unused epipen was also found in the drawer.";
cout<< " Rick had a lunch meeting at 1 pm on the same day with his new publisher, his manager
        and co-author.";
cout<< " The meeting was on the topic of signing a new book deal with the co-author.";
cout<< " All 3 had left the house at the same time.";
cout<< " Text messages on a phone owned by Rick were checked and it was seen that the manager
        had been urging Rick to not sign the deal.";
cout<< " Rick texted back later, saying it was too late now.\n\n";


cout<< " Detective, it is your duty to solve this high profile case as soon as possible. The media is
        at the door, and your preliminary investigation might just help to find who killed this world
        famous author.\n";
cout<< " Grab your magnifying glass, it is now time to investigate.\n";
getch();
```

```
clrscr();

        cout<< " What will you do now? \n";

        cout<<" 1. Question Rick's manager. His texts sound interesting.\n 2. Question Rick's co-author. What was their relationship like?\n 3. Question Rick's publicist. Did he play a part?\n 4. Question Stacy. What did she know?\n";

        cin>>b;

        getch();

        clrscr();


        if(b==1)

        {cout<< " You chose to interrogate Rick's manager.\n";

        cout<< " Select question:\n 1. Ask him about the texts.\n 2. Ask him about Rick.\n 3. Ask him about his relationship with Rick.\n 4. Ask him about the reply to the texts.\n";

        cin>>l;


        switch(l)

        {

        case 1:

        cout<<" The manager confesses he was worried about the deal between Rick and the co-author. He admits the deal wouldn't have been profittable for Rick. \n";

                score=score+20;

        break;

        case 2:

        cout<< "The manager says he cared for Rick a lot. He was aware about his routine, his allergies and everything in between. Rick had a severe peanut butter allergy.\n";

        score=score+20;
```

```
break;

case 3:

cout<<"His relationship with Rick extended beyond work. He cared for him and can't believe that he is dead.\n";

score=score+20;

break;

case 4:

cout<<" The manager was confused about the reply to the text. Rick had nothing to worry about. He was more famous than the co-author and could've easily refused to sign the deal.\n";

score=score+20;

default:

cout<< " Error ";

break;

}}


else if(b==2)

{cout<< " You decided to interrogate Rick's co-author \n";

cout<< " Select question:\n 1. Ask him about Rick.\n 2. Ask him about the manager's text to Rick.\n 3. Ask him about his relationship with Rick.\n 4. Ask him about the meeting.\n ";

cin>>l;

switch(l)

{

case 1:

cout<<" He says that Rick was an intersting person. Rick was more famous, and so this book deal would help the co-author a lot.\n";
```

```
score=score+40;

break;

case 2:

cout<<" The co-author was confused by the manager's texts. He thought everyone was happy with the deal as it would be beneficial for both of them. \n";

score=score+40;

break;

case 3:

cout<<" The relationship between the co-author and the Rick was a little strained. Rick had tried to use his fame to influence the deal. \n";

score=score+40;

break;

case 4:

cout<<" The meeting had not gone well. The manager had tried to drop the co-author from the deal. Rick had been silent the whole time.\n";

score=score+40;

break;

default:

cout<< " error \n";

break;

}}


else if(b==3)

{cout<< " You chose to interrogate Rick's publisher. \n ";

cout<< " Select question:\n 1. Ask him about the manager's texts to Rick.\n 2. Ask him about the meeting.\n 3. Ask him about the manager's relationship with Rick.\n 4. Ask him about Rick\n ";
```

```
cin>>l;

switch(l)

{

case 1:

cout<<" The publisher was aware about the manager's opinions on the deal. It did not come as a surprise to him.\n";

score=score+10;

break;

case 2:

cout<<"The publisher is new. He was mostly focused on taking notes. He did note the undercurrents of tension running through the meeting.\n";

score=score+10;

break;

case 3:

cout<<"The publisher is new. He is only aware that the manager and Rick were close.\n";

score=score+10;

break;

case 4:

cout<<"The publisher is new. He wasn't very close with Rick.\n";

score=score+10;

break;

default:

cout<< " error \n";

break;

}}
```

```
else

{cout<< " You chose to interrogate Stacy.\n ";

cout<< " Select question:\n 1. Ask her about Rick.\n 2. Ask her about the meeting.\n 3. Ask her about finding anything unusual.\n 4. Ask her about about the text messages.\n";

cin>>l;

switch(l)

{

case 1:

cout<<" Stacy was aware of Rick's daily routine. His eating like his peanut butter allergy for example, were known to her. \n";

score=score+30;

break;

case 2:

cout<<"She wasn't privy to the details of the meeting. She was just asked to cook lunch for the 4 of them.\n";

score=score+30;

break;

case 3:

cout<<"The fact that the dog was eating peanut butter was unusual. They didn't keep any peanut butter due to Rick's severe allergy. \n";

score=score+30;

break;

case 4:

cout<<"She wasn't aware about the text messages between Rick and his manager.\n";

score=score+30;

break;

default:
```

```
        cout<< " error \n";
        break;
        }}


getch();
clrscr();
cout<< " Choose who the culprit might be:\n 1. Manager\n 2. Publisher\n 3. Co-author\n 4.
        Stacy\n";


cin>>b;
if(b==1)
{cout<< "The manager was not the culprit. Try again.\n";
score=score-10;}
else if(b==2)
{cout<< "The publisher was not the culprit. Try again.\n";
score=score-10;}
else if(b==3)
{cout<< "Yes. It was the co-author who killed Rick.\n";
score=score+50;}
else if(b==4)
{cout<< "Stacy was not the culprit. Try again.\n";
score=score-10;}
else
{cout<< "Error";}
```

```
getch();

clrscr();

cout<< "\n\n\n\n\n\n\n\t\t\t Your final score is "<<score<<"\n\n\n\n\n";

cout<< "Score board:\n score<10 You might need to change your career.\n score=10 You need to
        do better than this. \n score=20 Your efforts were visible but today was not your day.\n
        score=30 You were on the right track.\n score=40 You knew what was happening all
        along.\n score=>50 Wow you deserve a promotion.";

getch();

clrscr();

cout<< "\n\n\n\n\n\n\n\n\n\n\n\t\t\t THANK YOU FOR PLAYING! \n\n\n\n\n";

getch();

clrscr();

return 0;

}
```

MODULES:

- **Module 1:**

Description: Header files and variable declaration.

Code:

```
#include<iostream.h>

#include<conio.h>

int main()

{clrscr();

int a, b, score=0, l, answer;

char Y, N, ans;
```

- **Module 2:**

Description: Code for title screen

Code:

```
cout<< "\n\n\n\n\n\n\n\n\n\n\t\t\t\t SLEUTH THE CROOK \n";

getch();

clrscr();
```

- **Module 3**

Description: Screen displaying ‘Are you ready to play’ with options ‘Yes’ and ‘No’

Code:

```
 cout<< "\n\n\n\n\n\n\n\n\t\t\t ARE YOU READY TO PLAY? \n ";

cout<< "\n\t\t Press Y to continue \t";

cout<< " Press N to leave \t";

cin>>ans;

clrscr();

if(ans=='N')
```

```
 {
 cout<< " Thank you for playing\n ";
 }
else
 {
 cout<< "\n\n\n\n\n\n\n\t\t\t\t Choose topic: \n ";
 cout<<"\n\t\t\t Case 1. The author's murder\n" ;
 cin>>a;
 clrscr();
 }
```

- **Module 4**

Description: Murder mystery storyline

Code:

```
cout<<" You chose case 1 \n";
        cout<< " Hello detective. Here is your case: \n\n";
cout<<" At 3:44 pm on a Sunday, first responders received a call that Rick, an author, was found dead in his living room.";
cout<< "There was no physical trauma to his body on a preliminary assessment.";
cout<< " The living room was found in shambles, with an upended sofa and stains on the rug.";
cout<< " 2 glasses with water in them were found on a nearby table.";
cout<< " His latest book was found torn near the table.";
cout<< " A broken vase laid near the window sill.";
cout<< " There was nobody else in the house, apart from his dog, which had been found eating peanut butter from the jar.";
cout<< " The call had been made by his househelp, Stacy, when she let herself in the house for cleaning.";
```

```
cout<< " An unused epipen was also found in the drawer.";

cout<< " Rick had a lunch meeting at 1 pm on the same day with his new publisher, his manager and co-author.";

cout<< " The meeting was on the topic of signing a new book deal with the co-author.";

cout<< " All 3 had left the house at the same time.";

cout<< " Text messages on a phone owned by Rick were checked and it was seen that the manager had been urging Rick to not sign the deal.";

cout<< " Rick texted back later, saying it was too late now.\n\n";


cout<< " Detective, it is your duty to solve this high profile case as soon as possible. The media is at the door, and your preliminary investigation might just help to find who killed this world famous author.\n";

cout<< " Grab your magnifying glass, it is now time to investigate.\n";

getch();

clrscr();
```

- **Module 5:**

Description: If-else conditional statement used for choosing character for interrogation. Nested switch case used to display statements as per user input.

Code:

```
    cout<< " What will you do now? \n";

    cout<<" 1. Question Rick's manager. His texts sound interesting.\n 2. Question Rick's co-author. What was their relationship like?\n 3. Question Rick's publicist. Did he play a part?\n 4. Question Stacy. What did she know?\n";

    cin>>b;

    getch();

    clrscr();


    if(b==1)
```

```
{cout<< " You chose to interrogate Rick's manager.\n";

cout<< " Select question:\n 1. Ask him about the texts.\n 2. Ask him about Rick.\n 3. Ask him about his relationship with Rick.\n 4. Ask him about the reply to the texts.\n";

cin>>l;


switch(l)

{

case 1:

cout<<" The manager confesses he was worried about the deal between Rick and the co-author. He admits the deal wouldn't have been profittable for Rick. \n";

        score=score+20;

break;

case 2:

cout<< "The manager says he cared for Rick a lot. He was aware about his routine, his allergies and everything in between. Rick had a severe peanut butter allergy.\n";

score=score+20;

break;

case 3:

cout<<"His relationship with Rick extended beyond work. He cared for him and can't believe that he is dead.\n";

score=score+20;

break;

case 4:

cout<<" The manager was confused about the reply to the text. Rick had nothing to worry about. He was more famous than the co-author and could've easily refused to sign the deal.\n";

score=score+20;

default:
```

```
cout<< " Error ";
break;
}}

else if(b==2)
{cout<< " You decided to interrogate Rick's co-author \n";
cout<< " Select question:\n 1. Ask him about Rick.\n 2. Ask him about the manager's text to Rick.\n 3. Ask him about his relationship with Rick.\n 4. Ask him about the meeting.\n ";
cin>>l;
switch(l)
{
case 1:
cout<<" He says that Rick was an intersting person. Rick was more famous, and so this book deal would help the co-author a lot.\n";
score=score+40;
break;
case 2:
cout<<" The co-author was confused by the manager's texts. He thought everyone was happy with the deal as it would be beneficial for both of them. \n";
score=score+40;
break;
case 3:
cout<<" The relationship between the co-author and the Rick was a little strained. Rick had tried to use his fame to influence the deal. \n";
score=score+40;
break;
case 4:
```

```
cout<<" The meeting had not gone well. The manager had tried to drop the co-author from the deal. Rick had been silent the whole time.\n";

score=score+40;

break;

default:

cout<< " error \n";

break;

}}


else if(b==3)

{cout<< " You chose to interrogate Rick's publisher. \n ";

cout<< " Select question:\n 1. Ask him about the manager's texts to Rick.\n 2. Ask him about the meeting.\n 3. Ask him about the manager's relationship with Rick.\n 4. Ask him about Rick\n ";

cin>>l;

switch(l)

{

case 1:

cout<<" The publisher was aware about the manager's opinions on the deal. It did not come as a surprise to him.\n";

score=score+10;

break;

case 2:

cout<<"The publisher is new. He was mostly focused on taking notes. He did note the undercurrents of tension running through the meeting.\n";

score=score+10;

break;
```

```
case 3:
cout<<"The publisher is new. He is only aware that the manager and Rick were close.\n";
score=score+10;
break;
case 4:
cout<<"The publisher is new. He wasn't very close with Rick.\n";
score=score+10;
break;
default:
cout<< " error \n";
break;
}}
else
{cout<< " You chose to interrogate Stacy.\n ";
cout<< " Select question:\n 1. Ask her about Rick.\n 2. Ask her about the meeting.\n 3. Ask her about finding anything unusual.\n 4. Ask her about about the text messages.\n";
cin>>l;
switch(l)
{
case 1:
cout<<" Stacy was aware of Rick's daily routine. His eating like his peanut butter allergy for example, were known to her. \n";
score=score+30;
break;
case 2:
```

```
cout<<"She wasn't privy to the details of the meeting. She was just asked to cook lunch for the 4 of them.\n";

score=score+30;

break;

case 3:

cout<<"The fact that the dog was eating peanut butter was unusual. They didn't keep any peanut butter due to Rick's severe allergy. \n";

score=score+30;

break;

case 4:

cout<<"She wasn't aware about the text messages between Rick and his manager.\n";

score=score+30;

break;

default:

cout<< " error \n";

break;

}}
```

- **Module 6**

Description: If-else conditional statements used to take user input and display result.

Code:

```
getch();

clrscr();

cout<< " Choose who the culprit might be:\n 1. Manager\n 2. Publisher\n 3. Co-author\n 4. Stacy\n";


cin>>b;
```

```
if(b==1)
{cout<< "The manager was not the culprit. Try again.\n";
score=score-10;}
else if(b==2)
{cout<< "The publisher was not the culprit. Try again.\n";
score=score-10;}
else if(b==3)
{cout<< "Yes. It was the co-author who killed Rick.\n";
score=score+50;}
else if(b==4)
{cout<< "Stacy was not the culprit. Try again.\n";
score=score-10;}
else
{cout<< "Error";}
```

- **Module 7**

Description: Code to display final score and score board

Code:

```
getch();
clrscr();
cout<< "\n\n\n\n\n\n\n\t\t\t Your final score is "<<score<<"\n\n\n\n\n";
cout<< "Score board:\n score<10 You might need to change your career.\n score=10 You need to do better than this. \n score=20 Your efforts were visible but today was not your day.\n score=30 You were on the right track.\n score=40 You knew what was happening all along.\n score=>50 Wow you deserve a promotion.";
```

- **Module 8**

Description: Code to display the ‘Thank you for playing’ screen.

Code:

```
getch();
clrscr();
cout<< "\n\n\n\n\n\n\n\n\n\n\n\t\t\t THANK YOU FOR PLAYING! \n\n\n\n\n";
getch();
clrscr();
return 0;
}
```

RESULT:

Fig 6: Title screen

Fig 7: 'Are you ready to play ?' screen for user input

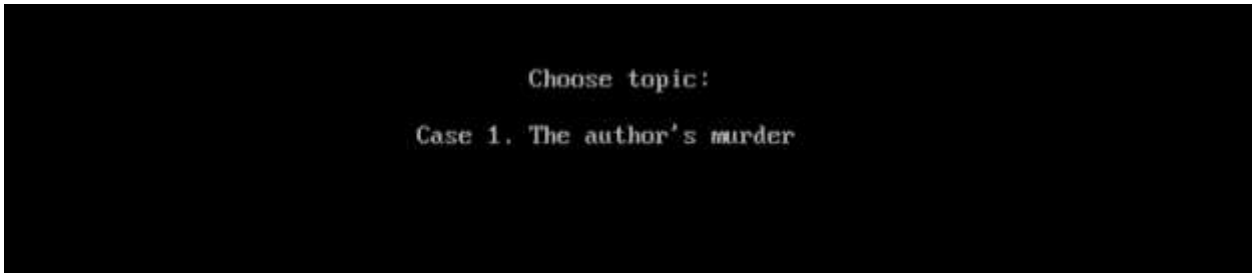

Fig 8: Name of the story

```
You chose case 1
Hello detective. Here is your case:

At 3:44 pm on a Sunday, first responders received a call that Rick, an author,
was found dead in his living room.There was no physical trauma to his body on a
preliminary assessment. The living room was found in shambles, with an upended s
ofa and stains on the rug. 2 glasses with water in them were found on a nearby t
able. His latest book was found torn near the table. A broken vase laid near the
 window sill. There was nobody else in the house, apart from his dog, which had
been found eating peanut butter from the jar. The call had been made by his hous
ehelp, Stacy, when she let herself in the house for cleaning. An unused epipen w
as also found in the drawer. Rick had a lunch meeting at 1 pm on the same day wi
th his new publisher, his manager and co-author. The meeting was on the topic of
 signing a new book deal with the co-author. All 3 had left the house at the sam
e time. Text messages on a phone owned by Rick were checked and it was seen that
 the manager had been urging Rick to not sign the deal. Rick texted back later,
saying it was too late now.

Detective, it is your duty to solve this high profile case as soon as possible.
The media is at the door, and your preliminary investigation might just help to
find who killed this world famous author.
Grab your magnifying glass, it is now time to investigate.
```

Fig 9: Output showing storyline

```
What will you do now?
1. Question Rick's manager. His texts sound interesting.
2. Question Rick's co-author. What was their relationship like?
3. Question Rick's publicist. Did he play a part?
4. Question Stacy. What did she know?
```

Fig 10: Output displaying options to choose character to interrogate by taking user input

```
What will you do now?
1. Question Rick's manager. His texts sound interesting.
2. Question Rick's co-author. What was their relationship like?
3. Question Rick's publicist. Did he play a part?
4. Question Stacy. What did she know?
1_
```

Fig 11: User input given as '1'

```
You chose to interrogate Rick's manager.
Select question:
1. Ask him about the texts.
2. Ask him about Rick.
3. Ask him about his relationship with Rick.
4. Ask him about the reply to the texts.
1_
```

Fig 12: Output displaying questions to be asked to character where user input is required

```
You chose to interrogate Rick's manager.
Select question:
1. Ask him about the texts.
2. Ask him about Rick.
3. Ask him about his relationship with Rick.
4. Ask him about the reply to the texts.
1
The manager confesses he was worried about the deal between Rick and the co-aut
hor. He admits the deal wouldn't have been profittable for Rick.
_
```

Fig 13: Output displaying statement on choosing question

```
Choose who the culprit might be:
1. Manager
2. Publisher
3. Co-author
4. Stacy
1_
```

Fig 14: Output to choose culprit by giving input

```
Choose who the culprit might be:
1. Manager
2. Publisher
3. Co-author
4. Stacy
1
The manager was not the culprit. Try again.
```

Fig 15: Output displaying the result

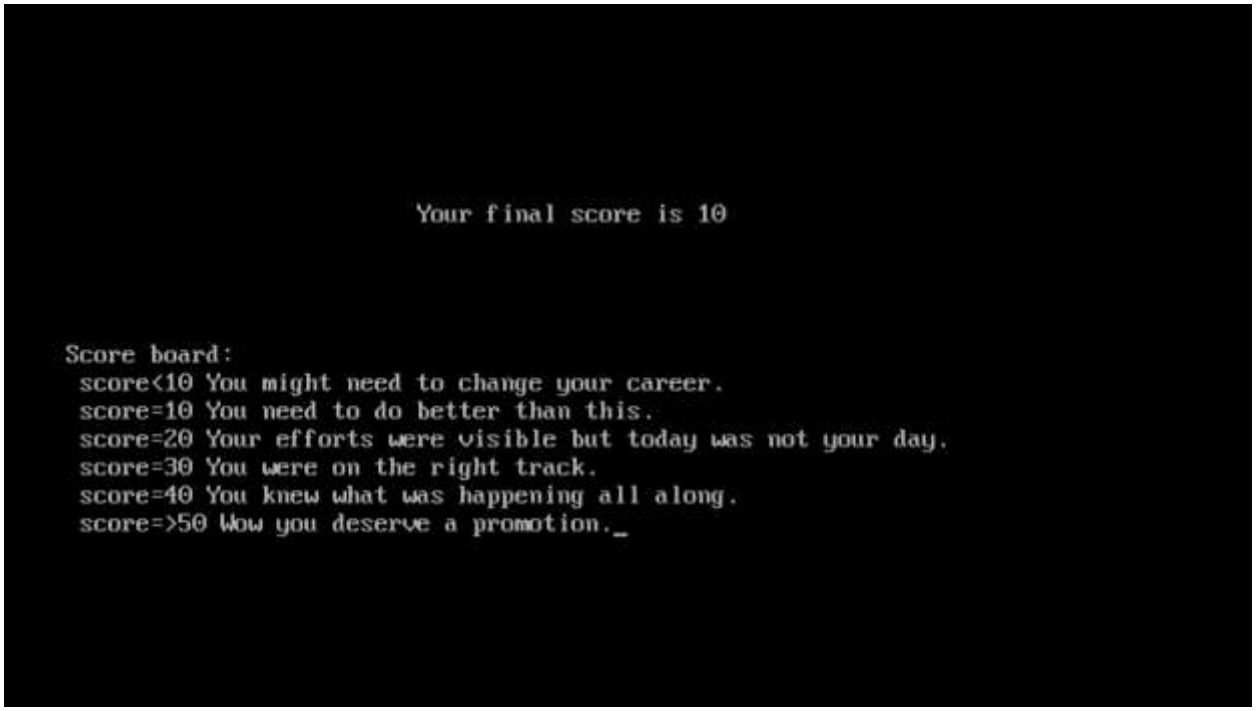

Fig 16: Output displaying the calculated score and the score board

Fig 17: Output displaying 'Thank you for playing' screen

CONCLUSION:

An interactive, murder mystery based story game was created using the programming language C++ with the help of software TurboC++. From the results obtained it can be concluded that the code was compiled successfully without any errors. A title screen with an aligned title was obtained. 'Are you ready to play?' screen with options 'Yes' and 'No' was displayed. A murder mystery storyline and questions based on the story were created. User input was taken and further questions were displayed as per the input of the user. The score was calculated after every question as per user input. The result was displayed as per user's input. The calculated score was displayed which depended on the choices that the user made. The score board was displayed below the calculated score and a 'Thank you for playing' screen was displayed, at the end of the game. The use of control structures such as if-else conditional statements and switch case were understood in detail through this project. This developed game can be included in computers, mobile phones and arcades, etc.

Aim, what was studied, obj completed (eg), where it can be used.

FUTURE SCOPE:

- Add graphics to the game to make it more interactive.
- Add more stories that vary by difficulty levels.
- Add a timer to the questions to make the game more challenging.
- Add a soundtrack to the game to make it more attractive.
- Change the background colour of the output.
- Add more questions to make the game more challenging.
- Add a screen to ask user whether they want to play again.
- Use the concepts of OOP to enhance the code.

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Aim

Obj

workflow

Benefits to the society

Conclude

Aim: To create an interactive, murder mystery based story game using the programming language C++ with the help of software TurboC++

Objectives:

- To create a title page which is visible first, on opening the game.
- To create a screen where user input is required to start the game (using if-else conditional statements)
- To create an original storyline based on a forensics murder mystery.
- To use if-else conditional statements for displaying different outputs based on user inputs.
- To use switch case nested within if-else conditional statements for displaying different outputs based on user inputs.
- To calculate score based on user input.
- To create a final result and create a score board to assess performance.
- To create a final 'Thank you for playing' screen.

Workflow:

Benefits to the society:

1. Boosted Creativity

In today's fast-paced, digital world, it's easy to fall into routine thinking. Storytelling games challenge you to think differently, to come up with unique solutions and narratives on the spot. They're like a workout for your creative muscles.

2. Enhanced Communication Skills

Clear and compelling communication is essential in every area of life – especially if you're an entrepreneur or aspiring speaker. Storytelling games give you a safe and enjoyable space to

practice these skills. You'll learn to organize your thoughts, craft engaging narratives, and respond effectively to unexpected twists.

3. Stress Relief

Taking a break and engaging in a creative activity can be a great way to unwind. It's like giving your brain permission to relax, play, and explore new ideas – without the pressure of work or deadlines.

Critical thinking
Players must navigate complex storylines, make decisions, and solve puzzles, which can help develop logical reasoning, creative thinking, and adaptability.

Immersion
Players can become invested in the game's narrative and feel emotionally attached to the characters.

FUTURE SCOPE:

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